JULIA L. SIMPSON, EIT

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SUMMARY STATEMENT

National Defense Science and Engineering Graduate (NDSEG) Fellow with expertise in climate and energy data analysis, machine learning, and scientific communication. Passionate about advancing climate resiliency; accelerating the energy transition; and integrating science and policy. 6+ years of combined research and engineering consulting experience.

EDUCATION

Columbia University, New York, NY
Master of Philosophy (M. Phil.) in Earth and Environmental Engineering. GPA 3.73 Expected Dec 2026

Master of Science (M.S.) in Earth & Environmental Engineering, GPA: 4.07 Feb 2024
Concentration in Sustainable Energy & Carbon with additional focuses on Data Science, Water Resources & Climate Risk

Washington University in St. Louis, St. Louis, MO
Bachelor of Science (B.S.) in Chemical Engineering, (Cum Laude), GPA: 3.73 May 2020
Minors in Energy Engineering, Environmental Engineering, and Psychology-Neuroscience-Philosophy

PROFESSIONAL EXPERIENCE

Graduate Researcher (Ph.D. Candidate), Columbia University, New York City, NY Aug 2022 - Present

- Applies spatial analysis to develop a risk-mitigation framework for wind energy along transmission corridors—where the same wind that drives generation also cools the line and raises its dynamic line rating (DLR), so transfer capacity peaks just as wind output does. Predicts wind speed between sparse measurement points, forecasts with calibrated confidence intervals, and culminates in sensor-placement recommendations to improve safety and reduce curtailment.
- Addresses oceanographic observational data scarcity in air-sea flux representation in climate modeling by developing a hybrid physics-machine learning workflow using ensembles of neural networks trained on 3 million synthetic samples, then fine-tuned via transfer learning to ~10,000 in-situ observations.
- Employs causal discovery methods (PCMCI+) to illuminate directional relationships between meteorological drivers and air-sea fluxes, enabling physically grounded interpretation of machine learning predictions.

Assistant Chemical Engineer, Burns & McDonnell, Wallingford, CT Aug 2020 - Dec 2022

- As Environmental Permitting Specialist: authored and submitted permit applications for energy infrastructure projects, including large-scale 55 and 65 megawatt (MW) solar installations, battery storage, and transmission lines.
- As Remediation & Air Emissions Scientist: analyzed regulatory frameworks to counsel clients on relevant requirements; executed field sampling campaigns, analyzed environmental data to evaluate remediation efficacy; and calculated facility potential to emit for 6 criteria and 188 hazardous air pollutants.

Sustainability Engineering Intern, Office of Sustainability, St. Louis, MO Jan 2020 - Mar 2020

- Collaborated on a team of 5 student interns to design 2.5 MW of solar installations for city of St. Louis. Developed a comprehensive proposal integrating SolSmart policy and Helioscope engineering design.

AWARDS

National Defense Science and Engineering Graduate Fellow Sep 2024 – Present
Earned a competitive three-year graduate fellowship from the Department of Defense, awarded through the Office of the Under Secretary for Research and Engineering and the military services (award rate under 6.3%).

"What We Do Matters" and "We Invest in a Better World" Burns & McDonnell, Wallingford, CT Jun 2022
Received two company culture recognitions for innovative solution to a project obstacle and dedication to a client.

TECHNICAL SKILLS

Python, Jupyter notebooks, Binder, Git, R, MATLAB, Gurobi optimization, Excel, CAD Civil3D, HYSYS, Helioscope, neural networks, machine learning, artificial intelligence, data science, climate data, public speaking & technical writing.